MIL-V-51316
6 May 1969

MILITARY SPECIFICATION

VANADIUM PENTOXIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of vanadium pentoxide (V_2O_5).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood-Preservative: Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Wood; Post Construction, 4-Way Entry.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Vanadium pentoxide shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as V_2O_5)	99.5	----	4.2.4.1
Chloride (Cl)	----	0.20	4.2.4.2

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the vanadium pentoxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitue a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 50 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Strapping or fiberboard packing aids incorrect.
(f) Wood preservation incorrect or missing (level A only).
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 50 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as V_2O_5). Weigh to the nearest 0.1 milligram (mg) approximately 0.4 gram (g) of the specimen into a 500-milliliter (ml) Erlenmeyer flask. Add 150 ml of water and then 30 ml of 1 to 1 sulfuric acid. Boil the solution on a hot plate for 5 minutes. Add 50 ml of water and continue to boil for a few minutes until a complete yellow solution is obtained. Bubble sulfur dioxide gas from a sulfur dioxide cylinder into the yellow solution for 10 minutes. The solution should now be a brilliant blue. Bubble carbon dioxide gas from a carbon dioxide cylinder into the blue solution while boiling gently on a hot plate. Continue bubbling the carbon dioxide gas for 30 minutes. While the solution is still hot (above 70° C), titrate the blue solution with 0.1N potassium permanganate solution to a faint tinge of yellow-orange endpoint. Run a blank determination in parallel with the specimen test. Calculate the percent vanadium pentoxide as follows:

$$\text{Percent vanadium pentoxide} = \frac{9.095\ A\ (B - C)}{M}$$

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where: A = Normality of potassium permanganate solution,
B = Milliliters of potassium permanganate solution used to titrate specimen,
C = Milliliters of potassium permanganate solution used to titrate blank, and
W = Weight of specimen in grams.

4.2.4.2 Chloride. Weigh 1.000 g of the specimen and dissolve in 1 to 1 nitric acid. Filter if necessary and add 1 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced in a control containing 2.0 mg of chloride ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound + 0.05 ounce of vanadium pentoxide shall be packaged in accordance with the applicable requirements of PPP-C-301 in a type I, class 1, widemouth glass bottle, placed in inclosure 6 as specified therein. The bottle shall also conform to glass type b, light penetration grade optional, with outer seal A band or tape of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of vanadium pentoxide, packaged as specified in 5.1.1, shall be intermediately packaged in accordance with the applicable requirements of PPP-C-301 except that table I therein shall not be used. The bottles shall be arranged in a three bottle by four bottle pattern. Fiberboard separators shall not be required.

5.1.2 Level C. One-quarter pound + 0.05 ounce of vanadium pentoxide shall be packaged as specified in 5.1.1 except that the bottle shall not be placed in inclosure 6 of PPP-C-301.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages, prepared as specified in 5.1.1.1, shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated-plywood box shall conform to type overseas, style I or J, for a type 2 load of PPP-B-601. The wirebound wood box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. The use of fiberboard fillers and pads shall be in accordance with PPP-C-301 to

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assure a tight pack. Each container shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. In addition, unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Packing shall be as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636. The box shall be closed in accordance with the requirements for taping style RSC boxes as exterior containers and then reinforced by tape banding as specified in the appendix to PPP-B-636. No preservative shall be required.

5.2.3 Level C. Twelve bottles of vanadium pentoxide, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade optional of PPP-B-636. The box shall be fitted with fiberboard half-slotted separators which shall form close fitting cells for each of the bottles and which shall be the same height as the bottles. All inner faces of each box shall be lined with full box-size fiberboard pads. Additional pads shall be added where needed to assure a tight fit. Fiberboard pads and separators shall conform to type CF, class domestic, variety SW, grade 125 of PPP-F-320. The box shall be closed as specified for closure of class domestic boxes in PPP-B-636.

5.3 Marking. In addition to the marking for unit containers specified in 5.3.1 and any marking required by the contract or order (see 6.2), unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis of maximum limit of chloride.
(f) Net weight of contents.
(g) Precautionary marking:

CAUTION!

HARMFUL IF INHALED

CAUSES EYE IRRITATION!

Use only with adequate ventilation.
Do not breathe vapor.

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5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using load type I except that strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness. The load tie-down straps parallel to the 48-inch pallet dimension shall be modified to serve as load unitizing straps, containing the load, but not the pallet. One load tie-down strap shall be added to, and centered between, those straps which are normally required to be parallel to the 40-inch pallet width. One horizontal strap shall be required for each tier of containers. The pallet shall conform to style 1, type optional, class 1, preserved of MIL-P-15011. All wood storage aids, if used, shall be preserved as specified for the pallet.

6. NOTES

6.1 Intended use. Vanadium pentoxide is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Kind of box for level A packing, if required (see 5.2.1).
(d) Additional marking, if required (see 5.3).
(e) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

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Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0457

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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